LAB EXPERIMENT RECORD

13-10-2026---Set 17: Employee Performance Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Day	TUESDAY

Aim

To analyze employee performance data using a line chart, bar chart and table, and to build an interactive Tableau dashboard for exploring employee performance.

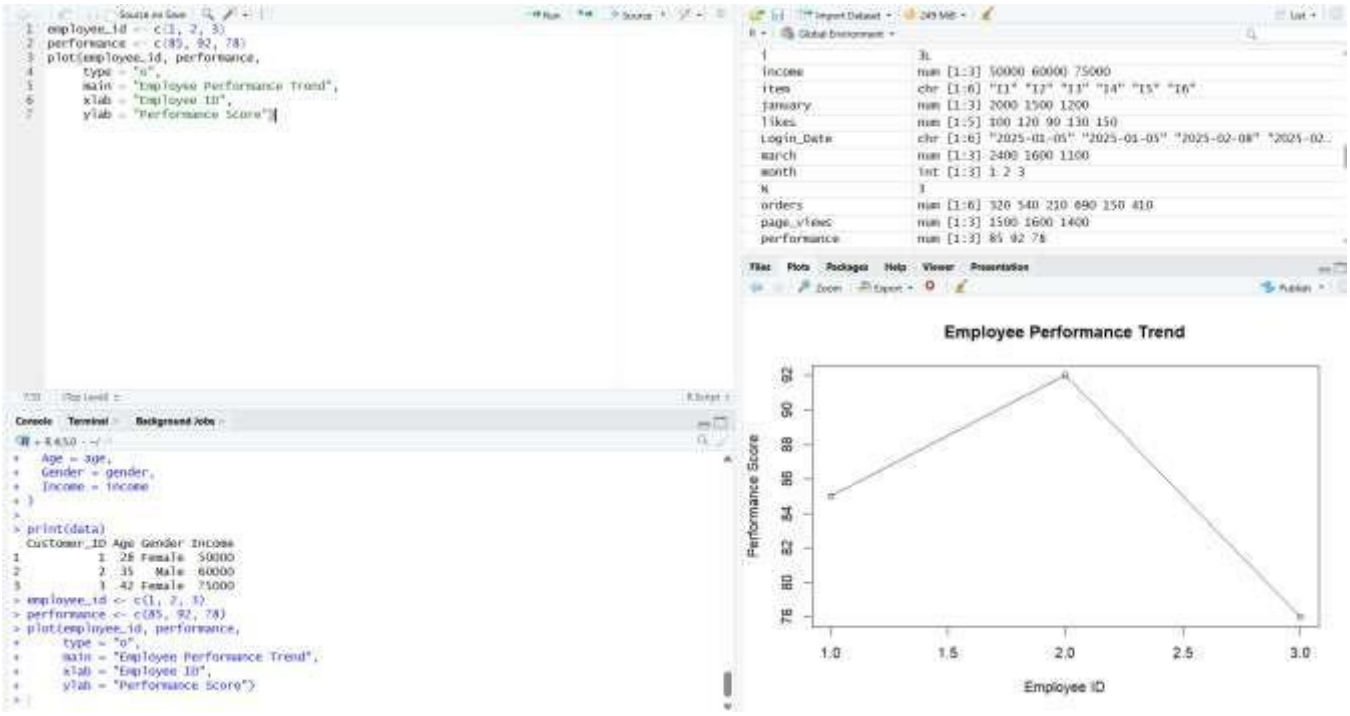
Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Question 1: Line Chart of Performance Trend

Question: Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

Aim: To visualize the performance trend of employees using a line chart with labelled axes and a title.

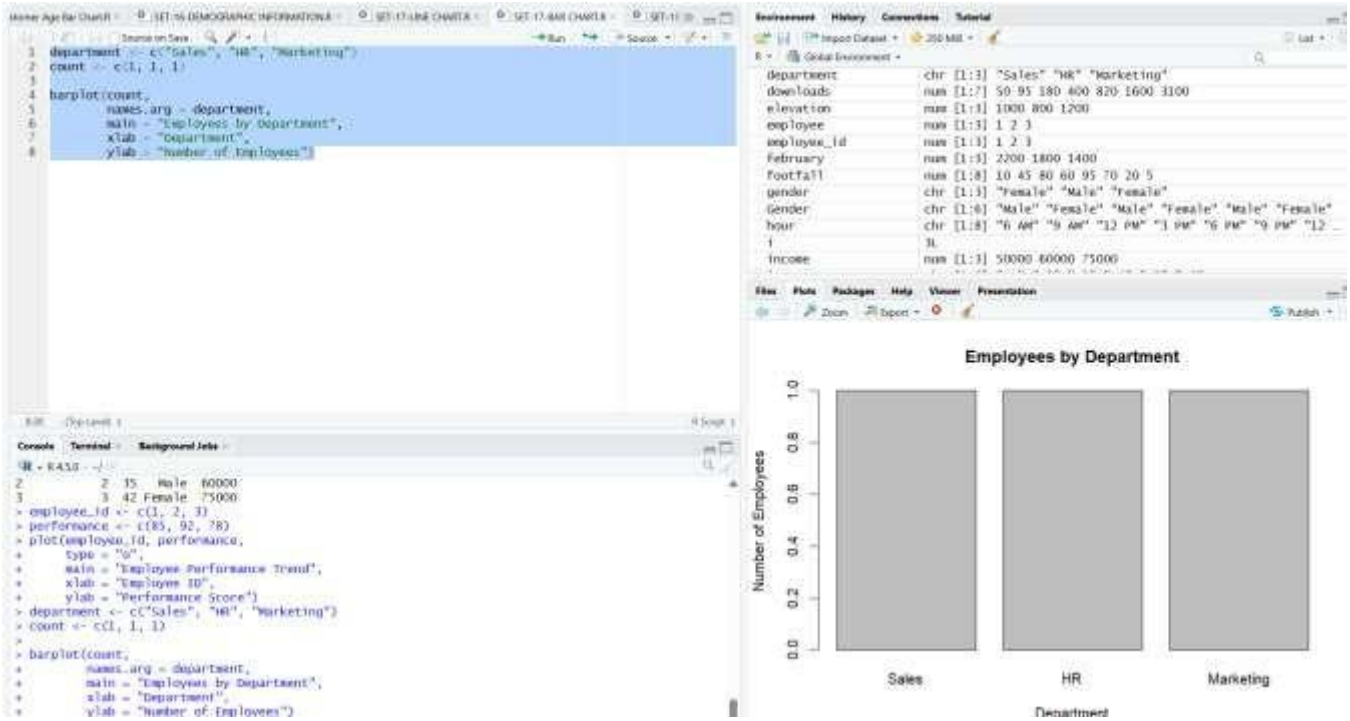


Result: Performance rises from 85 for Employee 1 to a peak of 92 for Employee 2, then falls to the lowest value of 78 for Employee 3.

Question 2: Bar Chart of Employees by Department

Question: Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

Aim: To show the distribution of employees across departments using a labelled bar chart.

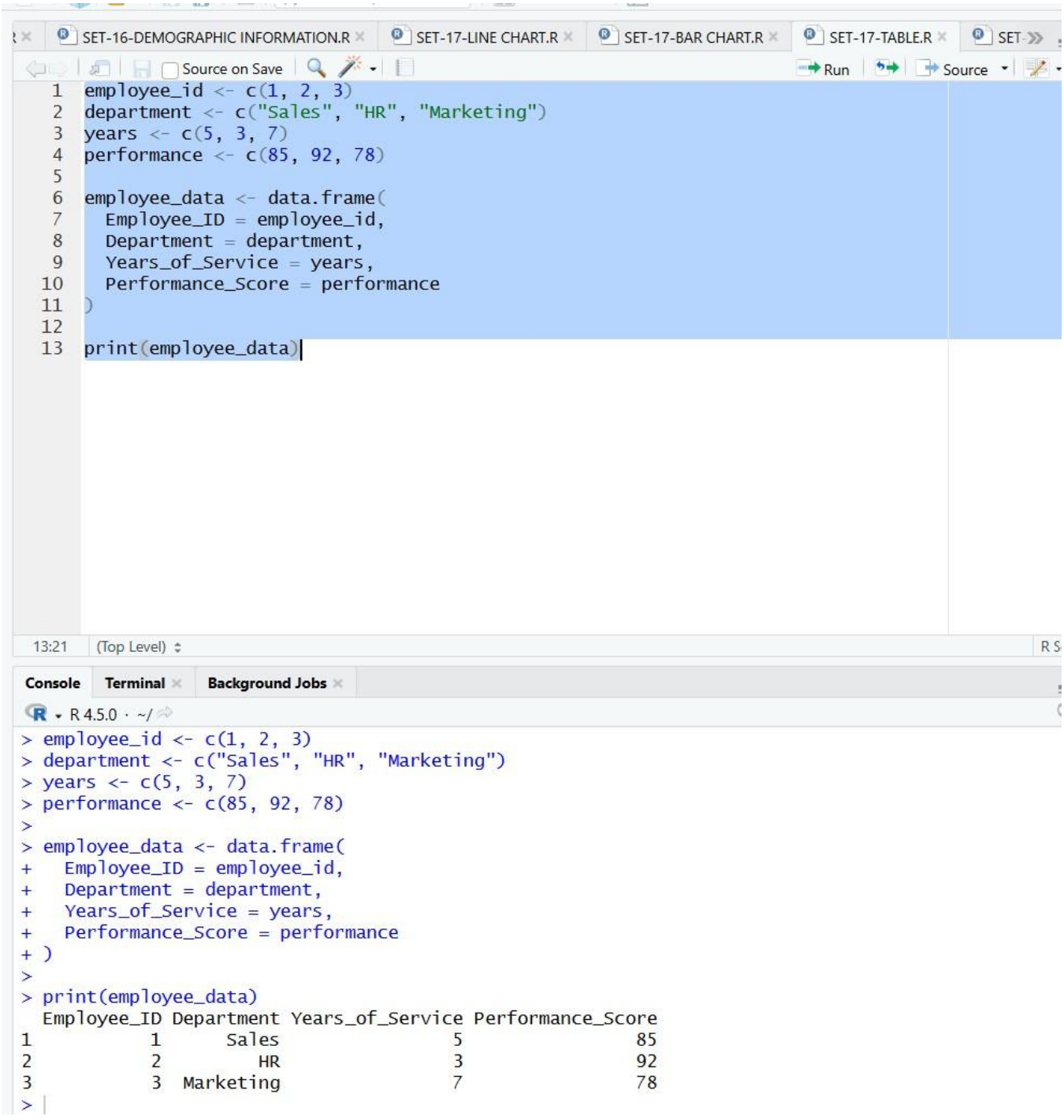


Result: Sales, HR and Marketing each have exactly 1 employee, so the three bars are equal in height.

Question 3: Table of Employee Performance

Question: Build a table to display the performance data for each employee. Label the table elements.

Aim: To present the department, years of service and performance score of each employee in a labelled table.



Result: The table lists all three employees; the HR employee has the highest score (92), and the Marketing employee, with the most service (7 years), has the lowest (78).

Question 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Aim: To build an interactive Tableau dashboard that combines the line chart, bar chart and table for exploring employee performance.



Employee ID	Name	Department	Years of Service	Performance Score
E001	John	HR	2	75
E002	Emma	Finance	4	82
E003	Liam	IT	1	68
E004	Olivia	Marketing	6	90
E005	Noah	Operations	3	78
E006	Ava	IT	5	85
E007	Lucas	Finance	2	72
E008	Mia	HR	4	88

Result: The dashboard places all three views together, and a department filter updates them so users can explore performance interactively.

Overall Result

Observation: The average performance score is 85, and a longer service period does not guarantee a higher score in this small sample.